A Rank-Three Haar Phase Law for a Four-Symbol Open Walsh Gate

Route-A structural certificate C168

Abstract

We study the natural four-symbol open Walsh gate $A = F_4^* \operatorname{diag}(1,0,1,1)$ at the complete register clock $C_k = A^{\otimes k}$. Its characteristic polynomial is $x(x-1)(x^2+ix/2-1/2)$, so the nonzero secular degree is exactly 3^k and the generalized zero space has dimension $4^k - 3^k$ for every register length. The two damped eigenvalues have normalized phases $u_\pm$ whose ratio r satisfies $r + r^{-1} = -3/2$. Since this nonintegral rational cannot be the algebraic-integer trace of a root of unity, r is non-torsion. The multiplicity-weighted phase measure is a three-step multiplicative walk, and its mth Fourier coefficient is $((1 + u_+^m + u_-^m)/3)^k$. Every fixed nonzero mode contracts strictly; hence the phase measures converge weakly to Haar measure on the circle. The one-site log modulus equals 0 with probability 1/3 and $-\log 2/2$ with probability 2/3. An exact mixed transform upgrades the marginal limits to a joint $N(0, (\log 2)^2/18) \otimes$ Haar law. The hole-zero control instead converges in total variation to the uniform four-phase group. A hole-reflection antiunitary identity changes both hole and propagation order; it does not construct a self-adjoint or target-spectrum operator.

Keywords: open quantum maps; Walsh dynamics; tensor spectra; phase equidistribution; Haar measure; algebraic integers.

中文摘要

本文研究自然四符号开放 Walsh 门 $A = F_4^* \operatorname{diag}(1,0,1,1)$ 在完整寄存器时钟 $C_k = A^{\otimes k}$ 下的精确谱结构。单点特征多项式为 $x(x-1)(x^2+ix/2-1/2)$，因此对任意寄存器长度，非零世俗多项式次数恰为 3^k，广义零特征空间维数为 $4^k - 3^k$。两个衰减特征值的单位相位记为 $u_\pm$，其比值 r 满足 $r + r^{-1} = -3/2$。若 r 是单位根，则该迹应为代数整数；但非整数有理数 $-3/2$ 不可能如此，故相位比非扭结。按代数重数加权后，相位测度成为三步乘法游走，其第 m 个傅里叶系数精确等于 $((1 + u_+^m + u_-^m)/3)^k$。每个固定非零模态均严格收缩，从而相位测度弱收敛到圆周上的 Haar 测度。单点对数模以概率 $1/3$ 取零、以概率 $2/3$ 取 $-\log 2/2$；精确混合变换进一步证明中心化涨落与相位联合收敛到方差为 $(\log 2)^2/18$ 的高斯分布和 Haar 测度的乘积律。零号孔洞对照则依总变差收敛到四相位有限群上的均匀律；孔洞反射的反酉恒等式同时改变孔洞与传播次序，不能被解释为固定孔洞自伴极限。结论只属于源侧开放动力学，不构造目标谱。

关键词：开放量子映射；Walsh 动力学；张量谱；相位等分布；Haar 测度；代数整数。

1 Frozen gate and one-site certificate

Let $F_4 = (i^{j\ell}/2)_{0 \le j,\ell < 4}$, and, for $h \in \{0,1,2,3\}$, let $P_h = \operatorname{diag}(p_0,p_1,p_2,p_3)$ with $p_j = 1 - \delta_{jh}$. Set $A_h = F_4^* P_h$ and write $A = A_1$. On pure tensors set

$$B_k(v_0 \otimes \cdots \otimes v_{k-1}) = v_1 \otimes \cdots \otimes v_{k-1} \otimes A v_0. \tag{T}$$

One application of B_k is one tick. In exactly k ticks every original factor passes through A once, hence one complete cycle is $C_k = B_k^k = A^{\otimes k}$.

Lemma 1. A is diagonalizable with eigenvalues

$$0, \quad 1, \quad \lambda_+ = \frac{\sqrt{7} - i}{4}, \quad \lambda_- = \frac{-\sqrt{7} - i}{4}. \tag{1}$$

For $u_\pm = \sqrt{2}\lambda_\pm$, the ratio $r = u_+/u_- = (-3 + i\sqrt{7})/4$ is not a root of unity.

Proof. Direct determinant expansion gives

$$\chi_A(x) = x(x-1)(x^2 + ix/2 - 1/2), \tag{2}$$

with four distinct roots (1), and $|\lambda_\pm| = 1/\sqrt{2}$. Exact radical arithmetic gives $r + r^{-1} = -3/2$. Were r a root of unity, this sum would be an algebraic integer. A rational algebraic integer is an integer, a contradiction. □

2 All-length secular and phase laws

For $a + b + c = k$, write $M_k(a,b,c) = k!/(a!b!c!)$.

Theorem 2. For every $k \ge 1$,

$$\det(I - zC_k) = \prod_{a+b+c=k} (1 - z\lambda_+^b \lambda_-^c)^{M_k(a,b,c)}. \tag{3}$$

The polynomial in (3) has degree 3^k, and the generalized zero eigenspace has dimension $4^k - 3^k$.

Proof. Tensor products of the four one-site eigenvectors form an eigenbasis. Nonzero products use the three nonzero roots; grouping words by their three contents gives (3) and the multinomial total 3^k. Every other tensor vector has a zero factor. $\square$

Let μ_k be the algebraic-multiplicity-weighted phase law of the nonzero spectrum. The same word count yields

$$\mu_k = 3^{-k} \sum_{a+b+c=k} M_k(a,b,c)\delta_{u_+^b u_-^c},$$

$$\widehat{\mu}_k(m) = \left(\frac{1+u_+^m+u_-^m}{3}\right)^k. \tag{4}$$

Theorem 3. μ_k converges weakly to normalized Haar measure on $\mathbb{T}$.

Proof. For $m \neq 0$, equality in the triangle inequality for the average in (4) would force $u_+^m = u_-^m = 1$, hence $r^m = 1$, contradicting Lemma 1. Every fixed nonzero Fourier mode therefore tends to zero. Density of trigonometric polynomials proves the claim. $\square$

The labels in (3) retain their multiplicities; distinctness across all triples is not asserted. Nor is a Fourier gap uniform in all modes claimed. At every finite k, μ_k is atomic, so its total-variation distance from continuous Haar measure is exactly one.

3 Joint modulus–phase theorem

Under the normalized nonzero spectral law, a tensor factor contributes $X = 0$ with probability $1/3$ and $X = -\log 2/2$ with probability $2/3$. Hence $\mathbb{E}X = -\log 2/3$ and $\operatorname{Var} X = (\log 2)^2/18$. Put $Y_k = (\log|\rho| + k\log 2/3)/\sqrt{k}$.

Theorem 4. For real t and integer m,

$$\begin{aligned}&\mathbb{E}[e^{itY_k}\,\mathrm{phase}(\rho)^m]\\ &\quad= e^{it\log 2\sqrt{k}/3}\left(\frac{1+e^{-it\log 2/(2\sqrt{k})}(u_+^m+u_-^m)}{3}\right)^k.\end{aligned} \tag{5}$$

Consequently $(Y_k, \mathrm{phase}(\rho)) \Rightarrow N(0, (\log 2)^2/18) \otimes \mathrm{Haar}$.

Proof. For $m = 0$, (5) is the centered iid characteristic function and converges to the stated Gaussian transform. For fixed $m \neq 0$, its base tends to the strict contraction in (4), so the mixed transform tends to zero. The scalar CLT supplies tightness; Fourier–characteristic uniqueness gives the product limit. $\square$

4 Torsion and antiunitary controls

For the already defined $A_0 = F_4^* \operatorname{diag}(0,1,1,1)$,

$$\chi_{A_0}(x) = x(x-1)(x+1/2)(x+i), \qquad \mathrm{phase}\,\sigma(A_0)\setminus\{0\} = \{1,-1,-i\}. \tag{6}$$

Let ν_k be the algebraic-multiplicity nonzero phase law of $A_0^{\otimes k}$, the k-fold convolution of the uniform law on $\{1,-1,-i\}$. Its nontrivial Fourier coefficients have modulus 3^{-k}, so finite Fourier inversion gives

$$\mathrm{TV}(\nu_k, \mathrm{Uniform}\langle i\rangle) \le \frac{3}{2}3^{-k}. \tag{7}$$

Thus the moved hole realizes a finite torsion law, not another Haar-circle limit. Let $R = F_4^2$, so $RA_1R = A_3$. If K is coordinate conjugation, the antiunitary $\Theta = F_4K$ obeys

$$\Theta A_1 \Theta^{-1} = A_3^* = \operatorname{diag}(1,1,1,0)F_4. \tag{8}$$

This exchanges holes 1 and 3 and reverses propagation/projector order. It supplies no fixed-hole self-adjoint or antiunitary limiting operator.

5 Evidence and boundary

The release freezes exact ledgers through $k = 24$ and modes $m = 24$. Independent strict Python and SymPy reconstructions, byte replay, and hostile semantic mutations test the implementation; the finite ledgers do not prove the all-k statements.

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